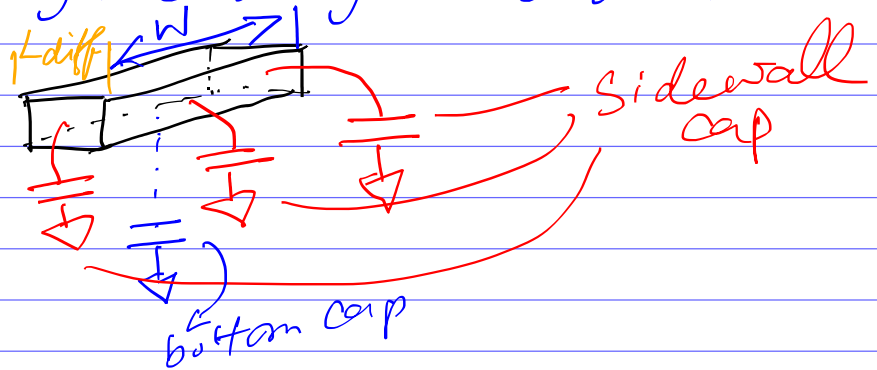


Inverter - Transient characteristics

Two sources of cap

(1) Junction cap of various p-n junctions

Consider the diffusion region (drain or source) surrounded by the body on all sides.

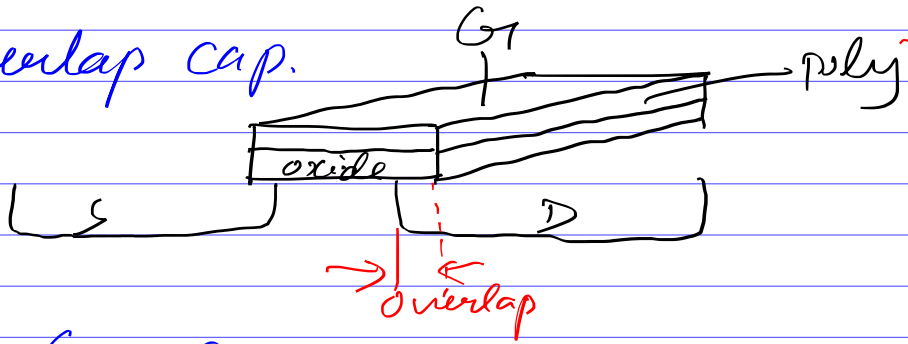


Cap. between drain/source and body

$$C_{DB_n} = C_{jsw}(2L_{diff} + W_n) + C_j \times W_n \times L_{diff} \rightarrow \text{bottom}$$

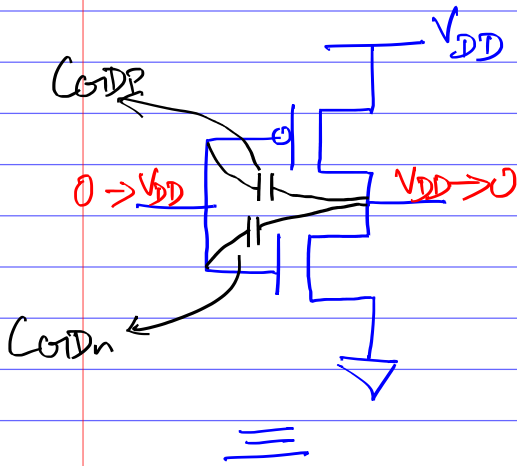
C_{jsw} - sidewall cap per unit length
 C_j - junction cap per unit area.

2.) Overlap cap.

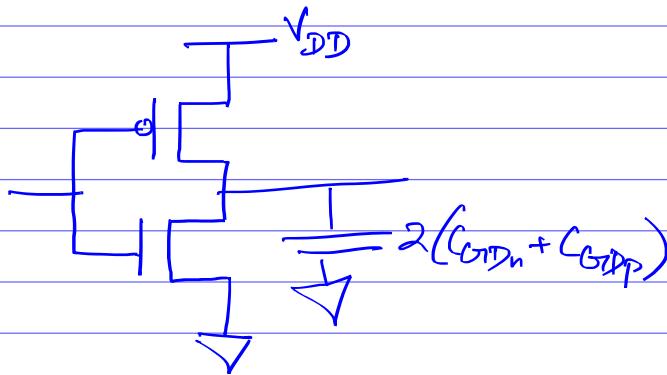


$$C_{GD}: C_{gd0} \times W_n$$

↳ overlap cap. per unit length.



When input switches from 0 → V_{DD}, output switches from V_{DD} → 0
 ⇒ the overlap cap sees a 2V_{DD} transition from a V_{DD} transition at input.



Total cap.

Assuming the inverter drives a load C_L , the total output cap.

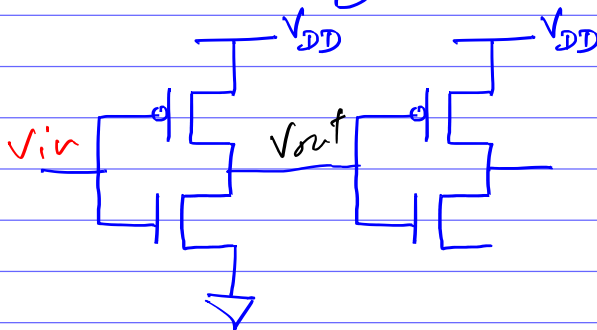
$$C_{\text{total}} = (C_{\text{ODN}} + C_{\text{ODP}}) \times 2 + C_{\text{DBn}} + C_{\text{DBp}} + C_L$$

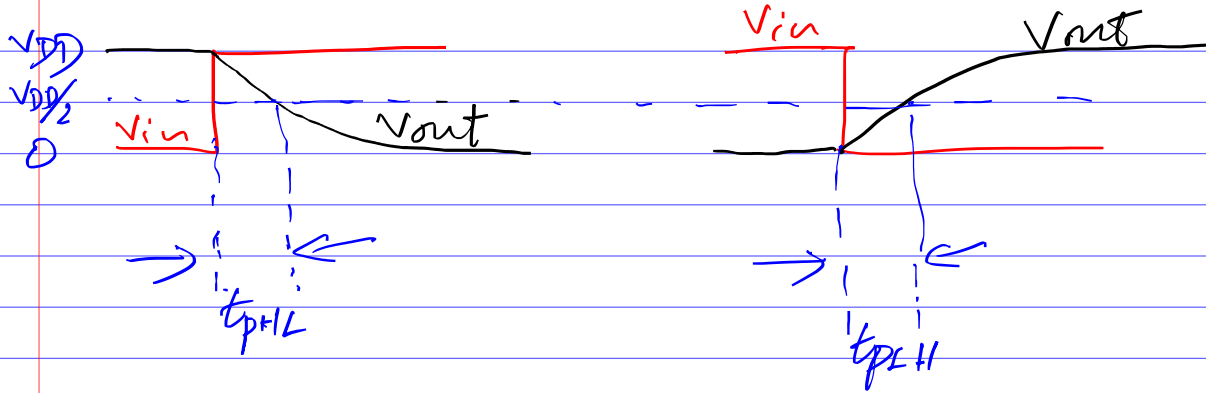
If the inverter drives an identical copy of itself the load cap C_L is the gate oxide caps. of p & nMOS.

$$C_L = C_{\text{oxn}} \times W_n \times L_n + C_{\text{oxp}} \times W_p \times L_p$$

Inverter delay

Consider an ideal transition at input of an inverter driving an identical copy of itself.





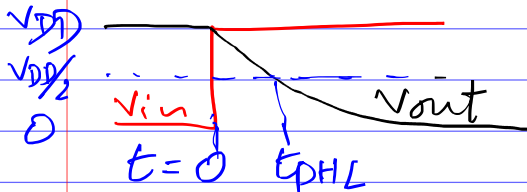
Delay is defined as the time between input $V_{DD}/2$ to output $V_{DD}/2$ for corresponding transitions.

t_{PHL} : delay for output to go from "High" to "Low"
 t_{PLH} : "Low" to "High"

$$t_p = (t_{PHL} + t_{PLH})/2$$

After the ideal input transition only nMOS or pMOS is ON to discharge or charge the output.

Let us analyse the output falling transition in detail. The same analysis holds good for rising transition replacing the nMOS expressions with pMOS.



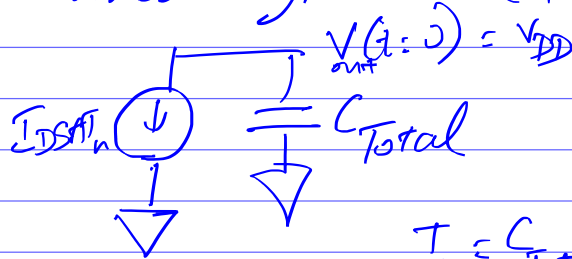
At $t=0$, $V_{in} = V_{out} = V_{DD}$
 $\Rightarrow$ nMOS is in sat.

At $t = t_{PHL}$, $V_{in} = V_{DD}$, $V_{out} = V_{DD}/2$

$$V_{DSATn} = \frac{(V_{GS} - V_{TH}) \times \mu_n C_{ox} L_n}{V_{GS} - V_{TH} + E C_{ox} L_n} \approx 0.5V$$

$\Rightarrow V_{out} > V_{DSATn} \Rightarrow$ nMOS is in sat.

The equivalent model for discharging the output load is a current source if we ignore channel length modulation.



$$I = C_{Tot} \frac{dV_{out}}{dt}$$

$$I_{DSATn} \int_0^{t_{PHL}} dt = C_{Tot} \times \frac{V_{DD}}{2}$$

$$\Rightarrow t_{PHL} = \frac{C_{Tot} V_{DD}}{2 I_{DSATn}}$$

$$C_{Total} = C_L + C_{intrinsic}$$

$$C_{\text{Total}} = C_{ox}(W_n L_n + W_p L_p) \} \approx 1.5 \text{ fF}$$

$$1.7 \text{ fF} \left\{ \begin{array}{l} G' W_n L_{\text{diff}} + C_j W_p L_{\text{diff}} \\ + \\ C_{jsw}(W_n + 2L_{\text{diff}}) + C_{jsw}(W_p + 2L_{\text{diff}}) \\ + \\ 2C_{gdo}(W_p + W_n) \end{array} \right.$$

$$C_{\text{Total}} = 3.2 \text{ fF} = \alpha_0 + \alpha_n W_n + \alpha_p W_p$$

For a symmetric P/N behaviour in DC transfer characteristic, $V_{th} = V_{DD}/2$.

Both n/p are in saturation for $V_{th} = V_{DD}/2$.

For no load condition, $I_{DSAT_n} = I_{DSAT_p}$ at $V_{th} = V_{DD}/2$

$$\begin{aligned} \mu_n C_{ox} \left(\frac{W}{L} \right)_n (V_{in} - V_{Tn})^2 \frac{E_c L_n}{V_{in} - V_{Tn} + E_c L_n} \\ = \mu_p C_{ox} \left(\frac{W}{L} \right)_p (V_{DD} - V_{in} - |V_{Tp}|)^2 E_c L_p \end{aligned}$$

$$\text{This gives } \left(\frac{W_p}{W_n} \right) \approx 2 \quad \frac{E_c L_p + V_{DD} - V_{in} - |V_{Tp}|}{E_c L_p + V_{DD} - V_{in} - |V_{Tp}|}$$

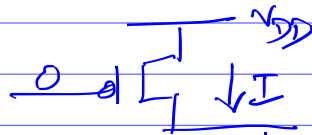
HW 1.) Calculate C_{Total} when an inverter with $(W_p/W_n) = 2$, $W_n = 0.42 \mu m$, $L = 0.15 \mu m$

$$C_{ox} = 8.4 fF/\mu m^2, C_j = 1.3 fF/\mu m^2, C_{jsw} = 0.238 fF/\mu m^2$$

$$C_{gd0} = 0.245 fF/\mu m^2$$

2.) Calculate t_{PHL} , t_{PLH} and t_p of the above inverter

Energy



$$E = \int_0^{\infty} V_{DD} \times I dt = V_{DD} \int_0^{\infty} C_{Total} \frac{dV_{out}}{dt} dt$$

$$= V_{DD} \int_0^{V_{DD}} C_{Total} dV_{out}$$

$$= C_{Total} V_{DD}^2$$

Power: If f_c is the switching frequency of clock. power = $f_c C_{Total} V_{DD}^2$.

Not all capacitances in the circuit switch in every cycle. $\alpha \in [0,1]$ is the activity factor. power = $\alpha C_{Total} V_{DD}^2 f_c$.

Power delay product: $C_{tot} V_{DD}^2 f_c \alpha \times t_p$
(PDP)

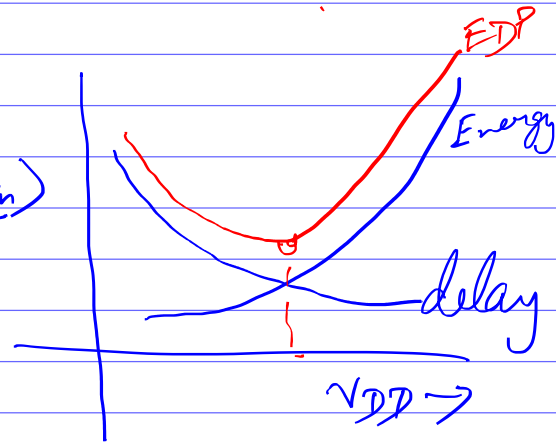
$$f_c < \frac{1}{2t_p} \Rightarrow C_{tot} V_{DD}^2 \frac{\alpha}{2}$$

Energy delay product: (EDP) $\hat{=}$ $\frac{C V_{DD}^2 t_p \times \alpha}{2}$

$$C_{tot} t_p \propto V_{DD} \frac{(E_{CL} + V_{DD} - V_{TN})}{(V_{DD} - V_{TN})^2}$$
$$\frac{C V_{DD}^2}{f_{DSAT}} \Rightarrow EDP \propto \frac{V_{DD}^3 (E_{CL} + V_{DD} - V_{TN})}{(V_{DD} - V_{TN})^2}$$

EDP is minimised when $\frac{\partial EDP}{\partial V_{DD}} = 0$

$$V_{DD} \approx 1.4V$$



Approximations made till here.

- 1.) C_{total} is a constant: it's a function of voltage depletion width changes with bias.
- 2.) V_{in} changes instantaneously: has a finite slope in reality.
↳ artefact is that p and n are "on" simultaneously

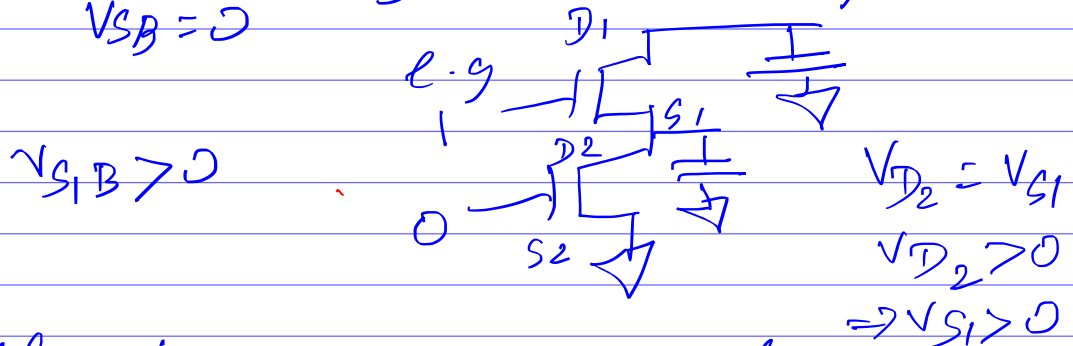
3.) Reverse bias leakage

4.) magical threshold voltage: subthreshold leakage

$$I_{DS} = I_{se} e^{\frac{qV_{GS}}{n k T}} \left(1 - e^{-\frac{qV_{DS}}{k T}} \right)$$

for $V_{GS} < V_{Tn}$

5.) Source and body are at same potential.
 $V_{SB} = 0$

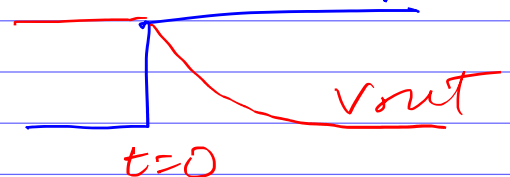
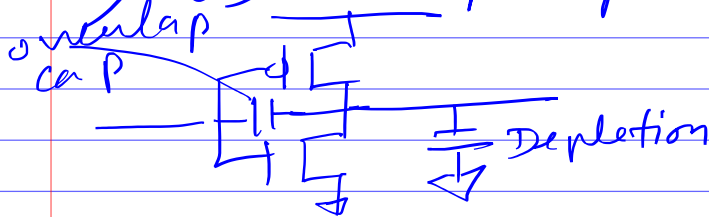


Threshold voltage of MOS is affected by V_{SB}

$$V_{Th} = V_{Th0} + \gamma \left(\sqrt{V_{SB} + |2\phi_F|} - \sqrt{|2\phi_F|} \right)$$

Body effect.

6.) C_{GD} overlap cap causes spikes at output.



At $t=0^-$, $V_{in}=0$, $V_{out}=V_{DD}$. C_{ov} is charged to $(-C_{ov}V_{DD})$

At $t=0^+$, $V_{in}=V_{DD}$. The charge in C_{ov} cannot discharge instantaneously since only nMOS is ON and it can only sink a max of I_{DSAT} .

To conserve the charge V_{out} has to go to $2V_{DD}$. The depletion caps resist this change to $2V_{DD}$ and hence the charge gets redistributed accordingly which causes a spike over V_{DD} whose peak is determined by the ratio of the overlap cap and the depletion cap.

